The effect of *Mycobacterium tuberculosis*-**infected** macrophages in the development of **lung cancer**

Salud
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INER

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1 INTRODUCTION

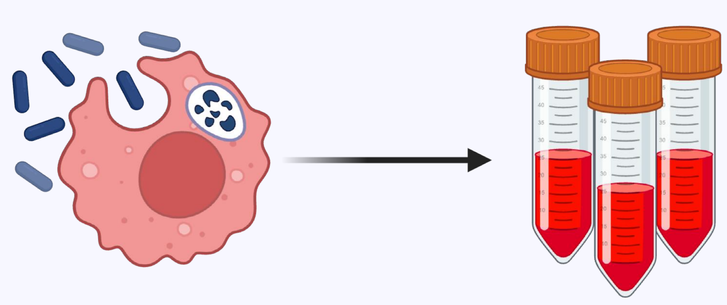
- Tuberculosis and lung cancer are two **major public health issues** worldwide
- Individuals with **tuberculosis are at increased risk** for developing lung cancer
- Tuberculosis-infected macrophages are the source of **pro-inflammatory cytokines** implicated in epithelial cell transformation.

2 OBJECTIVE

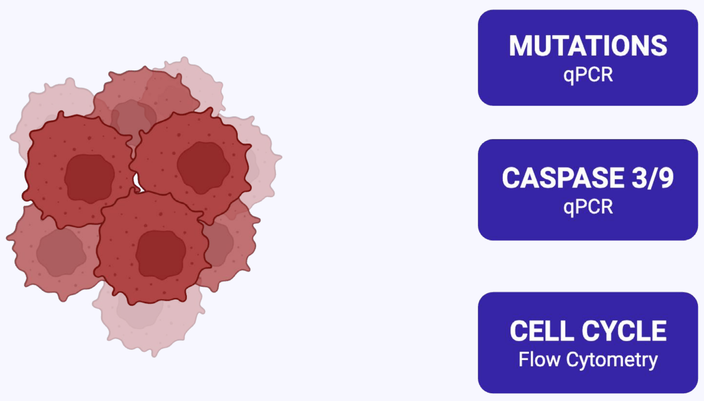
In the present work, we studied the contribution of the pro-inflammatory environment elicited by tuberculosis infected-macrophages in lung epithelial cells.

3 METHODS

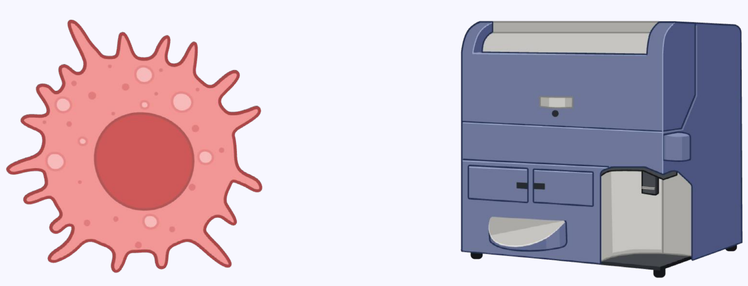
INFECTION OF THP-1-DERIVED MACROPHAGES WITH MTB H37Ra



EXPOSITION OF BEAS-2B CELLS TO CONDITIONED MEDIA



FLOW CYTOMETRY OF MTB-INFECTED THP-1 MACROPHAGES



4 RESULTS

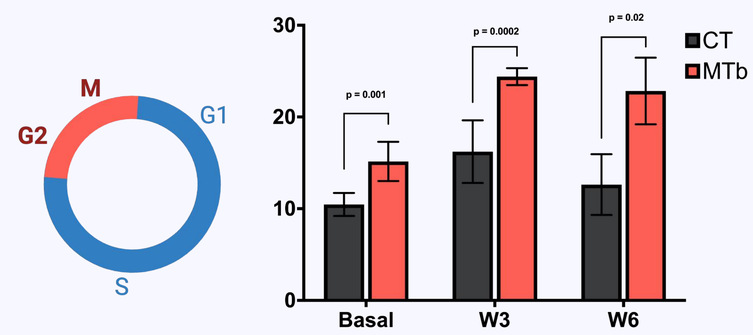


Figure 1. Conditioned media MTb-infected macrophages alters cell cycle of BEAS-2B cells. Percentage of cells in the G2/M phase are shown. Data represents mean ± SD.

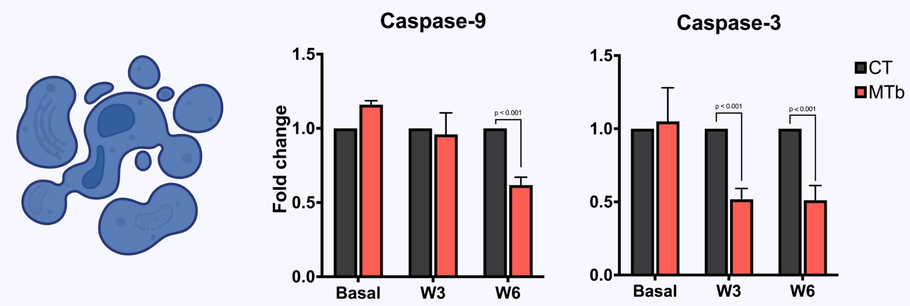


Figure 2. BEAS-2B cells decrease the expression of apoptotic molecules when exposed to conditioned media. Expression of *caspases 3* and *9* was quantified by qPCR. Data represents mean ± SD.

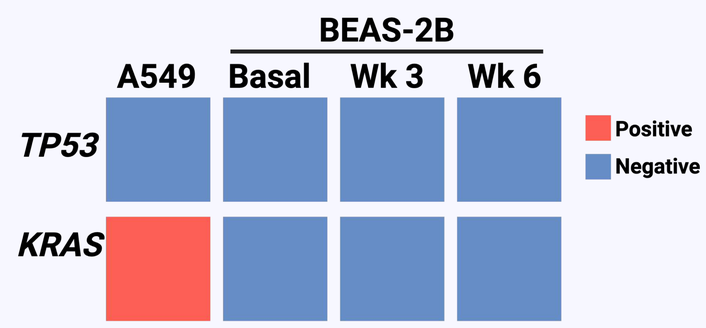


Figure 3. Evaluation of mutations in cancer driver genes in BEAS-2B cells after exposure to conditioned media. Mutations in TP53 (R273C) KRAS (G12C) genes were evaluated along treatment. A549 cell line was employed as control.

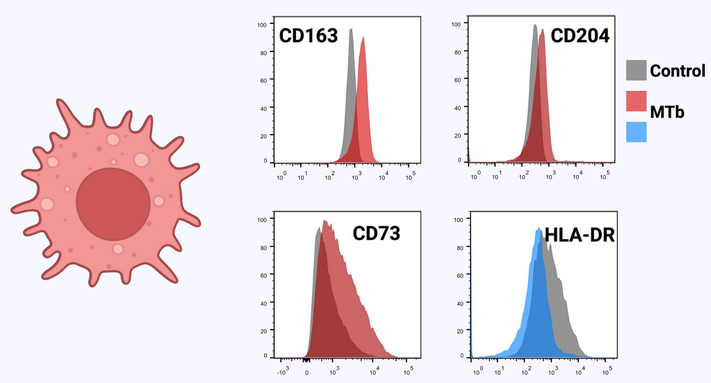


Figure 1. H37Ra-infected macrophages exhibit a M2 phenotype. The expression of M2 surface markers was evaluated in THP-1 CD14+ cells infected with MTb H37Ra at a MOI of 5:1.

5 PERSPECTIVES

- Continue studying the model
- Detecting and quantifying the expression of genes such as *Snai1, Twist1, Zeb, E-Cadherin,* and *N-Cadherin*
- Studying in human samples (biopsies) the presence of lung cancer molecules

6 CONCLUSIONS

Conditioned media from *Mycobacterium tuberculosis* H37Ra-infected macrophages **dysregulate the proliferation of lung epithelial cells as well as the expression of molecules implicated in cell death**.

REFERENCES

1.Global Tuberculosis Report 2024 (2024). 1 p.
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3.Nalbandian A, et al. *Oncogene* (2009) 28:1928–1938.